departure. You would have successfully used the black hole as a kind of time machine, allowing you to travel to earth's distant future.

To get a sense of the extreme scales involved, a star with the mass of the sun would be a black hole if its radius were not its actual value (about 450,000 miles), but, instead, just under 2 miles. Imagine: The whole of the sun squeezed to fit comfortably within upper Manhattan. A teaspoonful of such a compressed sun would weigh about as much as Mount Everest. To make a black hole out of the earth we would need to crush it into a sphere whose radius is less than half an inch. For a long time physicists were skeptical about whether such extreme configurations of matter could ever actually occur, and many thought that black holes were merely a reflection of an overworked theoretician's imagination.

Nevertheless, during the last decade, an increasingly convincing body of experimental evidence for the existence of black holes has accumulated. Of course, since they are black, they cannot be observed directly by scanning the sky with telescopes. Instead, astronomers search for black holes by seeking anomalous behavior of other more ordinary light-emitting stars that may be positioned just outside a black hole's event horizon. For instance, as dust and gas from the outer layers of nearby ordinary stars fall toward the event horizon of a black hole, they are accelerated to nearly the speed of light. At such speeds, friction within the maelstrom of downward-swirling material generates an enormous amount of heat, causing the dust-gas mixture to "glow," giving off both ordinary visible light and X rays. Since this radiation is produced just outside the event horizon, it can escape the black hole and travel through space to be observed and studied directly. General relativity makes detailed predictions about properties that such X ray emissions will have; observation of these predicted properties gives strong, albeit indirect, evidence for the existence of black holes. For example, mounting evidence indicates that there is a very massive black hole, some two and a half million times as massive as the sun, sitting in the center of our own Milky Way galaxy. And even this seemingly gargantuan black hole pales in comparison to what astronomers believe to reside in the core of the astonishingly luminous quasars that are scattered throughout the cosmos: black holes whose masses may well be billions of times that of the sun.